CT-KAE on transonic flow: reference and back001

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Models

Both are the same continuous-time Koopman autoencoder. A CNN encoder maps two input frames ($\Delta t = 0.1$) and the Mach number to $z \in \mathbb{R}^{256}$. The latent evolves exactly linearly, $z(\tau) = e^{\tau A(\mathrm{Ma})} z_0$, with A block-diagonal: 128 damped oscillators, decay $\gamma \geq 0$. A FiLM-conditioned CNN decoder maps z back to the four fields. 24.7 M parameters; trained 20k steps on the 33 training simulations to reconstruct and to predict 8 frames ahead, with latent consistency. **back001** adds backward latent consistency at weight 0.01. Three seeds each; every number below is a mean over seeds or a 3-seed average, as stated.

Main results

Table 1: Forecasting (MSE $\times 10^{-3}$, all frames; LSiM $\times 10^{-1}$), data assimilation (relative L_2 of the analysis at t_0; df1 headline: 12 problems, 2000 iterations; sweeps: mean over the points of six sweeps that every method was run on, 31 at $\delta_f = 1$ and 37 at $\delta_f = 9$), and inference cost (Tra_long, 240 steps, batch 1, RTX 4090). Lower is better throughout. Post-processing (MSE-optimal averaging) and the sharpened and LPM mirror averages (non-blurring) are described in the next two sections; the effect of post-processing on DA is in Table 6. Original KAE: final checkpoint; the published DA numbers used a checkpoint selected on the DA test set.

	MSE			LSiM			DA error			ms per
	ext	int	long	ext	int	long	df1 head.	df1 sweeps	df9 sweeps	step
ACDM	2.14	2.92	22.63	1.29	1.38	3.83	0.006	0.049	0.258	142.47
U-Net$_{ut}$	1.60	1.48	22.18	1.09	0.95	3.81	0.215	0.208	0.270	6.66
U-Net$_{tn}$	1.37	1.77	22.38	1.05	1.05	3.92	–	–	–	7.06
Original KAE (final ckpt)	2.05	11.05	18.47	–	–	–	0.029	0.029	0.103	0.80
Reference	1.91	3.37	16.08	1.41	1.71	4.06	0.029	0.046	0.095	0.27
Reference + post-proc.	1.31	2.45	11.87	1.73	1.86	7.24	–	–	–	7.68
Reference + sharp. mirror	1.34	2.57	15.81	1.20	1.53	4.07	–	–	–	1.59
Reference + LPM mirror	1.33	2.55	15.41	1.18	1.52	4.04	–	–	–	2.09
back001	2.03	3.04	15.11	1.49	1.71	4.88	0.024	0.031	0.088	0.27
back001 + post-proc.	1.46	2.46	11.97	1.90	1.81	7.47	–	–	–	7.68
back001 + sharp. mirror	1.53	2.59	14.88	1.33	1.57	4.91	–	–	–	1.59
back001 + LPM mirror	1.51	2.58	14.35	1.32	1.54	4.79	–	–	–	2.09

- **Long rollouts:** at 240 steps both CT-KAEs, without post-processing, have 27–33% lower MSE than ACDM and the U-Nets and 13–18% lower than the original KAE (46–47% and 35–36% with it), and stay correlated with the truth over 1000 steps (Fig. 3).
- **Data assimilation:** both are at least 2× more accurate than the U-Net and FNO at every lead. ACDM + SDA wins at $\delta_f \leq 2$. Against the original KAE, back001 is equal at short leads and better from $\delta_f = 8$; the reference is worse at short leads (Fig. 1).
- **Short horizon:** without post-processing the U-Nets are best on ext and int. With it the reference is best on ext; the U-Nets remain best on int. The LPM mirror average gives almost the same MSE without blurring: on ext the reference has lower MSE than every baseline (1.33) and better LSiM than ACDM (1.18 against 1.29); the U-Nets keep the best LSiM.
- **back001 vs reference:** back001 is better on DA, int and long, worse on ext. Post-processing removes most of the forecasting differences; the DA advantage remains.

- **Sharpness:** without post-processing the CT-KAE has worse LSiM than the baselines. Its spectrum matches the truth on ext and int and loses mid-band power on long rollouts (Table 10; an earlier version reported too-shallow spectra everywhere, an artefact of scoring unmasked fields). Post-processing improves MSE by blurring further (Table 3). The LPM and sharpened mirror averages do not blur: LSiM improves on ext and int, and on long it stays at or below the single decode's level (Table 4).

Post-processing

All steps act at inference on a trained model. Settings were fixed before the test sets were scored: mixture width and shrinkage on training rollouts, mirror averaging and the choice of steps on two proxy models trained with Mach 0.53–0.54 and 0.73–0.77 held out.

- **Mixture readout.** Average of 5 decodes of the same latent flow run at time scales $1 + \sigma\xi_q$ (Gauss–Hermite, $\sigma \approx 0.04$). Hedges the phase drift of the wake.
- **Mirror averaging.** The flow is (nearly) symmetric about the centre line: flip the input vertically with $v_y \to -v_y$, forecast, flip back, average with the plain forecast.
- **Climatology shrinkage.** Per lead and field, pull the forecast towards the Mach-interpolated training mean by a factor fitted on training rollouts. Affects long leads only.
- **Seed averaging.** Average the forecasts of three independently trained seeds.
- **Cost.** The full pipeline is 30 decodes per step: 7.7 ms per step against 0.27 for a single decode, about the cost of a U-Net and 18× cheaper than ACDM. Peak memory 0.9 GB.
- **Cost in sharpness.** Every step is an average, which MSE rewards and sharpness does not: LSiM and the spectral band errors get worse at every scale (Table 3). Use it when the target is MSE, not realistic fields.
- **Tested and rejected** on the proxies: test-time refinement of z_0 on the input frames (always worse), a mixture over neighbouring Mach numbers (no effect), lead-dependent mixture width, Wiener filtering by wavenumber (no better than one factor per lead and field).

Table 2: Each post-processing step added in turn: MSE $\times 10^{-3}$ (rows 1–4: mean over the 3 seeds; row 5: one 3-seed average), decoder calls per predicted step and wall clock per step.

		ext	int	long	decodes/step	ms/step
Reference	single decode	1.91	3.37	16.08	1	0.27
	+ mixture	1.77	3.16	13.03	5	1.28
	+ mirror	1.40	2.58	12.74	10	2.56
	+ shrinkage	1.40	2.58	11.98	10	2.56
	+ 3-seed average	1.31	2.45	11.87	30	7.68
back001	single decode	2.03	3.04	15.11	1	0.27
	+ mixture	1.88	2.89	13.45	5	1.28
	+ mirror	1.56	2.58	13.06	10	2.56
	+ shrinkage	1.56	2.57	12.07	10	2.56
	+ 3-seed average	1.46	2.46	11.97	30	7.68

Non-blurring post-processing

The aim is to lower the error without blurring. Two readouts qualify; both start from the mirror average of single decodes (no mixture) and restore what averaging removes.

- **Sharpened mirror average** (round 3). Rescale the anomaly of the average about the Mach-interpolated training mean, per frame, field and radial wavenumber, to the mean band power of the single decodes; phases kept, gain clipped to $[1, 5]$. One FFT pair per frame.
- **LPM mirror average** (round 4). Localized probability-matched mean (Ebert 2001; Clark 2017; Snook et al. 2020): in 16×16 patches (stride 8, Hann-weighted overlap) the fluid values of the average are replaced, rank for rank, by the pooled values of the single decodes. The average sets where structures are, the members set their amplitudes. No parameters fitted.

Table 3: What post-processing costs in sharpness. MSE $\times 10^{-3}$ on Tra_long; LSiM $\times 10^{-1}$; relative error of the $|v|$ spectrum on Tra_long (240 steps) in the low ($k \leq 4$), mid (5–12) and high (13–32) bands, and its mid-band slope; prediction and truth masked alike. Lower is better except the slope, which should match the ground truth.

	MSE	LSiM			spectrum, Tra_long			
	long	ext	int	long	low	mid	high	slope
ACDM	22.63	1.29	1.38	3.83	0.06	0.11	0.04	-3.30
U-Net$_{ut}$	22.18	1.09	0.95	3.81	0.05	0.08	0.03	-3.41
Reference	16.08	1.41	1.71	4.06	0.17	0.35	0.21	-3.44
Reference + post-proc.	11.87	1.73	1.86	7.24	0.32	0.61	0.51	-3.35
Reference + sharp. mirror	15.81	1.20	1.53	4.07	0.18	0.37	0.23	-3.55
Reference + LPM mirror	15.41	1.18	1.52	4.04	0.20	0.39	0.27	-3.47
back001	15.11	1.49	1.71	4.88	0.17	0.40	0.25	-3.53
back001 + post-proc.	11.97	1.90	1.81	7.47	0.31	0.62	0.50	-3.37
back001 + sharp. mirror	14.88	1.33	1.57	4.91	0.18	0.38	0.22	-3.59
back001 + LPM mirror	14.35	1.32	1.54	4.79	0.20	0.40	0.26	-3.52
Ground truth								-3.32

- **Selection.** Every candidate was scored on the proxies' held-out Mach numbers (7 simulations; 28 windows of 58 leads and 7 of 238) with a rule fixed before scoring: LSiM no worse than the single decode and band power at most 0.05 dex lower, on both proxies and both horizons; then lowest MSE. Round 3 had no admissible candidate (the sharpened mirror average missed only on the reference proxy at 238 leads, LSiM $+0.07 \pm 0.05$). In round 4 the LPM mirror average is the only admissible candidate: MSE -10% and LSiM -0.12 at 58 leads, MSE -3 to 0% and LSiM -0.09 to -0.23 at 238 leads. The proxies have one seed, so seed averaging is not validated there.
- **Test** (Table 4; per seed, paired against the single decode, SE over 3 seeds). LPM: on ext and int MSE falls by 11–23% and LSiM by 0.12–0.18; on long MSE falls by 3% (2–3 SE) and LSiM by 0.01–0.08 (within 1.5 SE). It is the only readout that lowers both at every horizon. The sharpened average has slightly higher MSE and, on long, LSiM +0.05, but keeps the high band better: ≤ 0.015 dex from the single decode against 0.01–0.025 dex for LPM, and a high-band spectral error of 0.07–0.09 against 0.10–0.12 on ext and int (Table 10). Over 1000 steps the mean v_x correlation is 0.716 / 0.713 (reference / back001) with LPM, 0.705 / 0.699 sharpened, 0.691 / 0.682 for the single decode and 0.730 / 0.722 with post-processing (Fig. 3). Late in that rollout, where the members disagree, LPM leaves patch-scale speckle in the far wake (frames 750 and 997).
- **Against post-processing** (3 seeds, reference): LPM gives MSE 1.33 / 2.55 / 15.41 against 1.31 / 2.45 / 11.87 (ext / int / long), LSiM 1.18 / 1.52 / 4.04 against 1.73 / 1.86 / 7.24. Use the LPM mirror average when realistic fields matter. The long-horizon MSE gain of post-processing comes only from blurring.

Table 4: Non-blurring readouts, each step added in turn. MSE $\times 10^{-3}$, LSiM $\times 10^{-1}$, and the mean $\log_{10}$ power ratio (prediction / truth) of the fluid-masked $|v|$ spectrum in the high band (13–32; 0 is exact, negative is smoother). Step rows: mean over 3 seeds, or one 3-seed average; *vs single*: paired change of the per-seed readout against the single decode, ± SE over seeds. LPM replaces sharpening; it is not applied on top.

		MSE			LSiM			high-band log ratio			decodes	ms
		ext	int	long	ext	int	long	ext	int	long	/step	/step
Reference	single decode	1.91	3.37	16.08	1.41	1.71	4.06	-0.03	-0.03	-0.11	1	0.27
	+ mirror	1.44	2.66	15.20	1.33	1.65	4.34	-0.06	-0.08	-0.16	2	0.52
	+ sharpen	1.48	2.74	15.82	1.25	1.57	4.11	-0.03	-0.04	-0.12	2	0.55
	+ 3-seed average	1.34	2.57	15.81	1.20	1.53	4.07	-0.03	-0.04	-0.12	6	1.59
	sharpen vs single	-0.43 ± 0.11	-0.64 ± 0.05	-0.26 ± 0.25	-0.16 ± 0.00	-0.14 ± 0.01	$+0.05 \pm 0.04$	$+0.002 \pm 0.002$	-0.015 ± 0.001	-0.008 ± 0.002		
	+ mirror, LPM	1.47	2.72	15.61	1.24	1.56	4.05	-0.04	-0.05	-0.13	2	0.76
	+ 3-seed average	1.33	2.55	15.41	1.18	1.52	4.04	-0.05	-0.06	-0.14	6	2.09
	LPM vs single	-0.44 ± 0.11	-0.66 ± 0.04	-0.47 ± 0.23	-0.18 ± 0.00	-0.15 ± 0.01	-0.01 ± 0.03	-0.010 ± 0.003	-0.025 ± 0.002	-0.019 ± 0.002		
back001	single decode	2.03	3.04	15.11	1.49	1.71	4.88	-0.03	-0.03	-0.12	1	0.27
	+ mirror	1.61	2.67	14.24	1.47	1.64	5.26	-0.06	-0.06	-0.16	2	0.52
	+ sharpen	1.66	2.72	14.92	1.37	1.61	4.93	-0.03	-0.04	-0.12	2	0.55
	+ 3-seed average	1.53	2.59	14.88	1.33	1.57	4.91	-0.03	-0.04	-0.11	6	1.59
	sharpen vs single	-0.37 ± 0.05	-0.32 ± 0.09	-0.19 ± 0.10	-0.12 ± 0.02	-0.10 ± 0.02	$+0.05 \pm 0.05$	-0.001 ± 0.001	-0.009 ± 0.002	-0.004 ± 0.007		
	+ mirror, LPM	1.65	2.71	14.66	1.37	1.59	4.80	-0.04	-0.05	-0.13	2	0.76
	+ 3-seed average	1.51	2.58	14.35	1.32	1.54	4.79	-0.04	-0.05	-0.13	6	2.09
	LPM vs single	-0.38 ± 0.05	-0.33 ± 0.09	-0.45 ± 0.14	-0.12 ± 0.02	-0.12 ± 0.02	-0.08 ± 0.05	-0.012 ± 0.001	-0.019 ± 0.002	-0.012 ± 0.004		

Literature coverage. Table 5 lists the post-processing families tried, from NWP statistical post-processing, ensemble summaries, spectral and stochastic methods, and generative refinement (the survey, with references, is `results/postproc3/literature_survey.md`). All were scored like the two readouts above. The perception–distortion theorem (Blau & Michaeli 2018) bounds what any of them can do: no estimator has both the MSE of

the posterior mean and the true spectrum; MSE-fitted combinations reach low MSE by blurring, and methods that restore the truth's spectrum pay in MSE (Freirich et al. 2021).

Table 5: Post-processing families tested. Proxy: change against the single decode on the held-out Mach numbers at 58 / 238 leads (MSE ratio; LSiM change $\times 10^{-1}$), mean of the two proxies. Fitted weights use leave-one-simulation-out cross-validation.

Family	References	Tested as	Result
Averaging (ensemble mean, test-time augmentation)	Puny et al. 2022; Shanmugam et al. 2021	mixture, mirror, seeds (adopted post-processing)	MSE 0.89 / 0.93 (mixture + mirror); blurs, LSiM +0.08 / +1.7
MOS, superensemble, per-band stacking	Glahn & Lowry 1972; Krishnamurti et al. 1999; Bowler et al. 2006	least-squares weights on 4 members, one per field or per field and wavenumber, 3 lead bins	MSE 0.85–0.86 / 0.90–0.92; LSiM +0.18–0.20 / +1.2–1.5: blurs; re-sharpened, the MSE gain is lost
Member-by-member blend	Van Schaeybroeck & Vannitsem 2015	$\frac{1}{2}$(mixture + mirror) + $\frac{1}{2}$ single	MSE 0.89 / 0.93; LSiM +0.02 / +0.75
Spectral rescaling (perception–distortion transport)	Blau & Michaeli 2018; Freirich et al. 2021	to the single-decode power (sharpen); partial transport, $\alpha \in [0.25, 1]$, to the truth's training power	sharpen: selected in round 3; full transport to the truth: MSE +22 to +46%; half transport of the mirror average: best long LSiM (−0.34) at MSE +12%
Probability matching	Ebert 2001; Clark 2017; Snook et al. 2019, 2020	global quantile mapping; LPM of mirror, of mixture + mirror, of the sharpened or stacked average	global: high band still lost; **LPM mirror: selected**; LPM after sharpening: best long LSiM, slightly less MSE gain
Field alignment, morphing	Ravela et al. 2007; Beezley & Mandel 2008; Lee et al. 2025	dense optical flow (Farneback) aligned averages; integer x-shift	no better than the plain average
Scale-selective blending, spectral nudging	von Storch et al. 2000; Seed 2003	mixture + mirror for $k \leq 2$ or 4, sharpened mirror above	no better than sharpening; $k \leq 4$ blurs on long
Fourier amplitude averaging	–	mean amplitude, phase of the mean	slightly worse than sharpening
Stochastic backscatter	Leith 1990; Shutts 2005	random-phase noise up to the single or true band power	no gain at best; MSE +13 to +110% when the mixture average is filled up
Latent averaging	–	average of the mixture nodes' latents	equals the single decode at 58 leads, blurs at 238
Generative refinement (SDEdit)	Meng et al. 2022; Shu et al. 2023	trained ACDM, noise level $t_0 \in \{1, \ldots, 12\}$ of 20, one sample, on test	ext, int: MSE −1 to −3%, LSiM +0.02 to −0.12; long: MSE +0 to +2%; $t_0 \leq 5$ doubles the high-band error; (t_0+1) ACDM calls per frame

Not tested. Need new training: learned combiners with sharpness-aware losses (AMSE, Subich et al. 2025; Grönquist et al. 2021), residual or corrective diffusion (CorrDiff, Mardani et al. 2025), posterior-mean rectified flow (Ohayon et al. 2025), GAN correction (Harris et al. 2022). Not a point forecast: ensemble copula coupling and the Schaake shuffle (Schefzik et al. 2013; Clark et al. 2004). Marginal distributions only: QDM, MBCn (Cannon et al. 2015; Cannon 2018). Analogs (Delle Monache et al. 2013) and Wasserstein barycenters (Agueh & Carlier 2011): no route to lower MSE than a member for signed velocity fields.

In data assimilation only mirror averaging applies: 4D-Var is run once in the plain frame and once in the mirrored frame, and the two analyses are averaged. It cuts the analysis error by 10–16% for both models at twice the cost (Table 6). Re-mapping the averaged analysis by LPM onto the two member analyses keeps almost all of the gain (−9 to −15% against −10 to −16%); it is 0.5–2% worse than the plain mirror average at every lead δ_f.

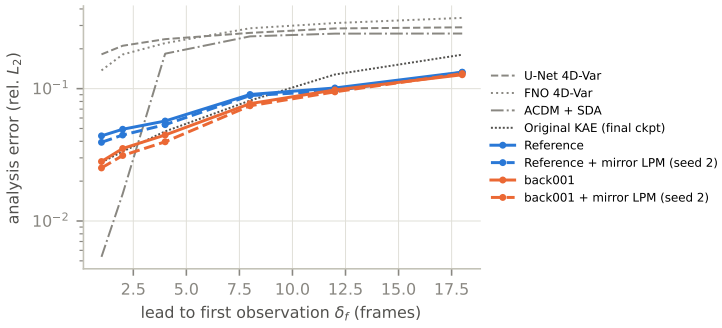


Figure 1: Analysis error against the lead δ_f to the first observation, transonic flow, $\delta_l = 25$, 5 observations, 500 iterations, 8 problems per point.

Table 6: Mirror averaging in 4D-Var: analysis error, seed 2 of each model, same problems as Table 1.

		df1 head.	df1 sweeps	df9 sweeps
Reference	plain	0.030	0.047	0.098
	mirror	0.026	0.042	0.088
	change	-13%	-10%	-10%
	mirror + LPM	0.026	0.043	0.089
	change	-12%	-10%	-9%
back001	plain	0.024	0.032	0.092
	mirror	0.021	0.027	0.082
	change	-16%	-15%	-11%
	mirror + LPM	0.021	0.027	0.083
	change	-15%	-14%	-10%

The paper's transonic experiments, re-run

Table 7: MSE ($\times 10^{-3}$, all fields) at $t = 15, 30, 45, 60$ of the 60-step rollouts. CT-KAE rows without post-processing: mean over 3 seeds. Best per column in bold.

	Tra_ext				Tra_int			
t	15	30	45	60	15	30	45	60
ACDM	0.40	1.36	3.51	6.51	0.43	2.07	5.13	8.46
U-Net$_{ut}$	0.14	0.92	2.70	5.14	0.17	0.96	**2.60**	**4.90**
U-Net$_{tn}$	**0.12**	**0.74**	2.44	4.50	**0.12**	**0.82**	3.00	6.87
Original KAE	0.44	1.10	2.42	3.92	1.27	5.30	9.75	15.78
Reference	0.36	1.54	3.17	4.84	0.64	2.90	5.58	8.48
Reference + post-proc.	0.25	1.05	**2.18**	3.32	0.49	2.20	4.11	5.64
Reference + sharp. mirror	0.25	1.08	2.24	3.34	0.49	2.22	4.29	6.02
Reference + LPM mirror	0.25	1.08	2.21	**3.30**	0.50	2.23	4.28	5.95
back001	0.36	1.64	3.36	5.30	0.60	2.64	5.10	7.48
back001 + post-proc.	0.25	1.14	2.44	3.79	0.47	2.20	4.21	5.73
back001 + sharp. mirror	0.25	1.14	2.54	4.00	0.47	2.23	4.45	6.22
back001 + LPM mirror	0.25	1.15	2.50	3.93	0.47	2.23	4.43	6.16

Short horizon. The U-Nets are best up to $t = 30$. On ext the post-processed reference is best from $t = 45$. On int the U-Net$_{ut}$ stays best; at $t = 60$ the post-processed CT-KAEs are second (5.6–5.7 against 4.9), ahead of ACDM (8.5) and the original KAE (15.8).

Table 8: The latent ODE $\dot{y} = Ay$ integrated numerically instead of $e^{\tau A}$ (seed 0, single decode). MSE $\times 10^{-3}$.

	integrator	ext	int	long
Reference	$e^{\tau A}$ (exact)	1.61	3.32	16.66
Reference	RK4, $\Delta t = 0.1$	1.61	3.32	16.66
Reference	RK4, $\Delta t = 0.3$	1.63	3.28	16.34
Reference	RK4, $\Delta t = 0.5$	1.92	3.22	14.58
Reference	midpoint, $\Delta t = 0.1$	1.77	3.49	17.10
Reference	Euler, $\Delta t = 0.05$	2.28	3.76	24.82
Reference	Euler, $\Delta t = 0.1$	6.94	6.11	diverged
Reference	Dopri5 (adaptive)	1.61	3.32	16.66
back001	$e^{\tau A}$ (exact)	1.98	2.91	14.57
back001	RK4, $\Delta t = 0.1$	1.98	2.91	14.57
back001	RK4, $\Delta t = 0.3$	2.00	2.90	14.46
back001	RK4, $\Delta t = 0.5$	2.29	2.97	14.17
back001	midpoint, $\Delta t = 0.1$	2.02	3.01	14.71
back001	Euler, $\Delta t = 0.05$	2.61	3.71	diverged
back001	Euler, $\Delta t = 0.1$	8.81	12.37	diverged
back001	Dopri5 (adaptive)	1.98	2.91	14.57

Table 9: Unseen time step: MSE at leads on the training grid (0.1 s) and at the half-steps between them (0.05 s, never trained on). Mean over 3 seeds.

	ext		int	
	0.1	0.05	0.1	0.05
Reference	1.98	1.99	3.49	3.51
back001	2.10	2.11	3.14	3.16

Table 10: Spectrum of $|v|$: relative error in the low ($k \leq 4$), mid (5–12) and high (13–32) wavenumber bands and the fitted mid-band slope. Prediction and truth are both multiplied by the fluid mask, as for MSE and LSiM. Without it the CT-KAE fields, which carry the normalisation mean inside the cylinder, show a spurious step at its edge; the earlier version of this table was computed that way.

	Tra_int (60 steps)				Tra_long (240 steps)			
	low	mid	high	slope	low	mid	high	slope
ACDM	0.03	0.02	0.03	-3.32	0.06	0.11	0.04	-3.30
U-Net$_{ut}$	0.02	0.02	0.02	-3.33	0.05	0.08	0.03	-3.41
FNO$_{16}$	0.06	0.09	0.18	-3.41	0.64	0.50	0.40	-3.64
Reference	0.04	0.04	0.06	-3.34	0.17	0.35	0.21	-3.44
Reference + post-proc.	0.06	0.17	0.25	-3.43	0.32	0.61	0.51	-3.35
Reference + sharp. mirror	0.03	0.04	0.09	-3.35	0.18	0.37	0.23	-3.55
Reference + LPM mirror	0.04	0.06	0.12	-3.36	0.20	0.39	0.27	-3.47
back001	0.04	0.06	0.07	-3.40	0.17	0.40	0.25	-3.53
back001 + post-proc.	0.06	0.16	0.23	-3.46	0.31	0.62	0.50	-3.37
back001 + sharp. mirror	0.04	0.06	0.08	-3.40	0.18	0.38	0.22	-3.59
back001 + LPM mirror	0.04	0.07	0.11	-3.40	0.20	0.40	0.26	-3.52
Ground truth				-3.33				-3.32

Findings.

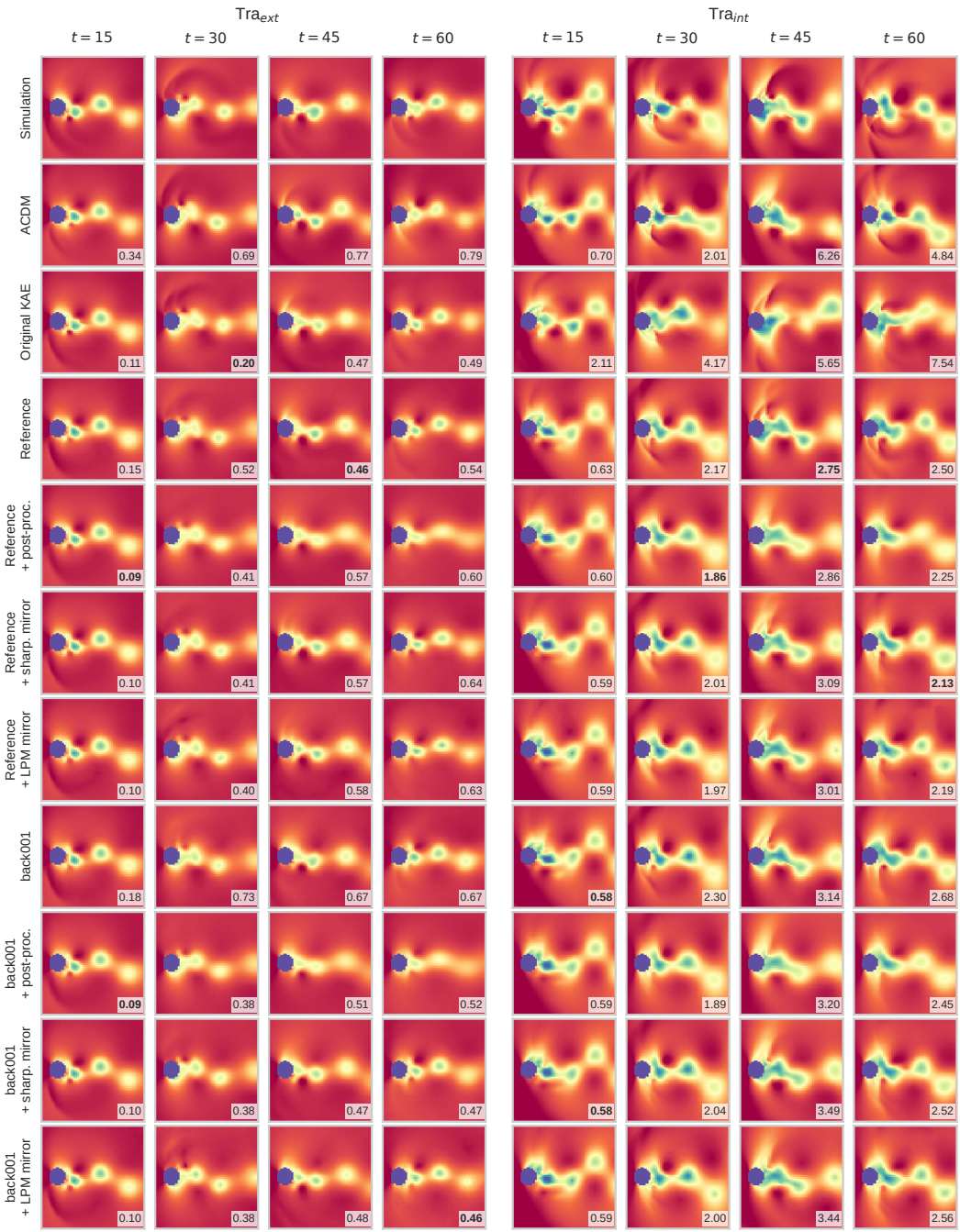


Figure 2: Pressure at $t = 15, 30, 45, 60$, first sequence of each test set (paper layout). CT-KAE rows without post-processing: seed 0; post-processing, sharpened and LPM mirror: 3-seed average. Number in each panel: pressure MSE ($\times 10^{-3}$) of that frame over the whole domain; bold is the best model in the column.

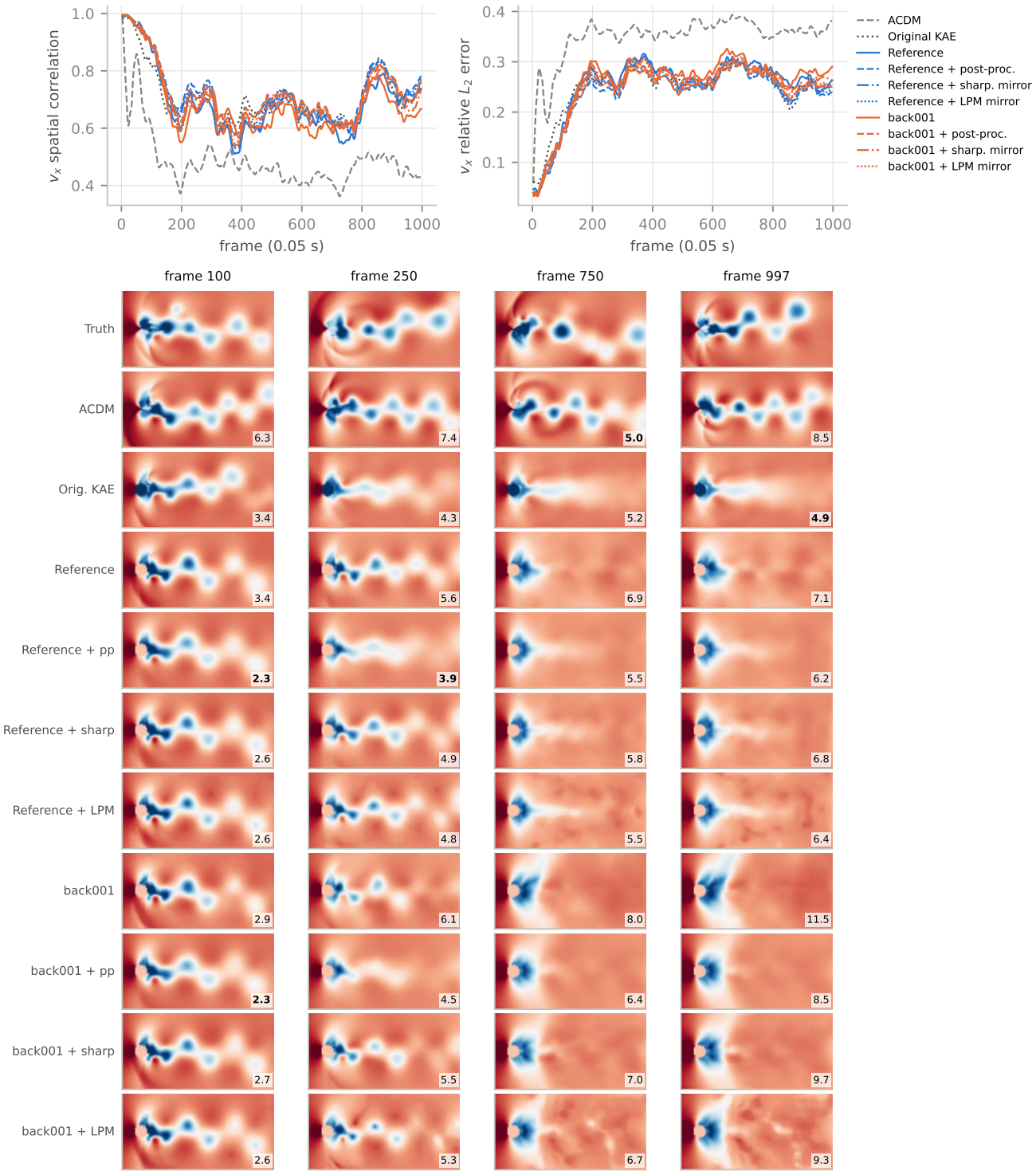


Figure 3: 1000-step rollout (Tra_long simulations, 0.05 s per frame, 2 trajectories). Top: v_x spatial correlation and relative L_2 error per frame. Bottom: pressure, first trajectory. Post-processing here is the 3-seed average of mixture + mirror, without shrinkage; "sharp" and "LPM" are the 3-seed sharpened and LPM mirror averages. Number in each panel: pressure MSE ($\times 10^{-3}$) of that frame; bold is the best in the column.

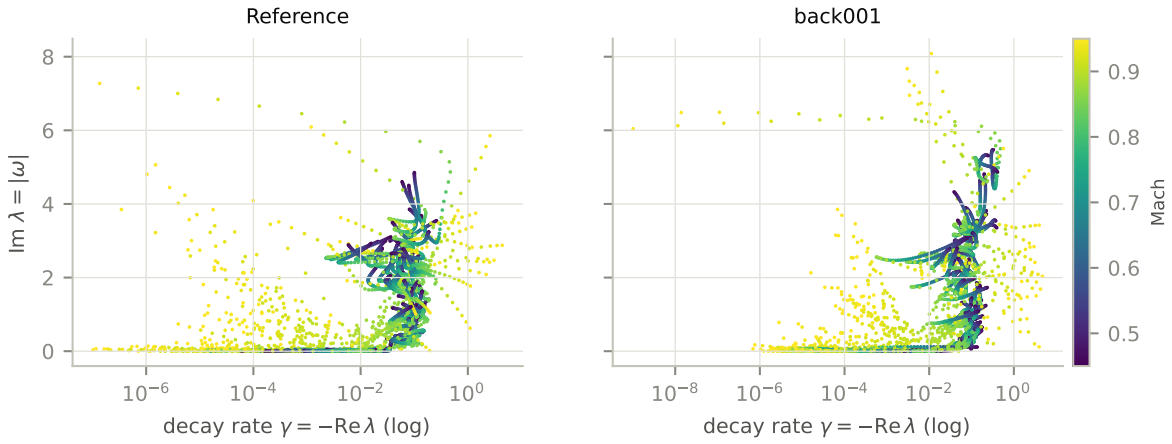


Figure 4: Generator eigenvalues $\lambda = -\gamma \pm i\omega$ for Mach 0.45–0.95 (seed 0).

- **Integrators.** RK4 matches the exact exponential up to $\Delta t = 0.3$; explicit Euler diverges on long rollouts (from $\Delta t = 0.05$ for back001), as the oscillators are almost undamped.
- **Unseen time step.** Forecasts between the training time steps are as accurate as on them (Table 9).
- **Spectra.** On Tra_int the CT-KAE slope matches the truth (-3.34 against -3.33) and its band errors are about 1.5–3× those of ACDM and the U-Net, well below FNO's. On Tra_long the mid-band error is 0.35–0.40 against 0.08–0.11 and grows with the lead: the CT-KAE loses small-scale structure late in the rollout. Post-processing roughly doubles or triples the band errors. The sharpened mirror average raises them by at most 10%, except int's high band (+50%); LPM raises the high-band error by 6–100% (int 0.12 against 0.06), a third or less of what post-processing adds. The CT-KAE's MSE advantage on long rollouts comes from tracking phase, not from resolving small scales.
- **Not re-run:** the original KAE's loss and LoRA ablations need retraining; the CT-KAE's own ablation study (in the full report) replaces them.